

Nima Aghayan

Senior Embedded Software Engineer | Automotive Software | Embedded Linux

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PROFESSIONAL SUMMARY

Embedded Software Engineer with 5+ years of experience in automotive software, Embedded Linux, telematics, and modem systems. Strong hands-on background in Embedded C, Yocto, Linux services, CAN, cellular connectivity, OTA/FOTA, eCall, and low-level drivers. Proven across the development lifecycle from requirements and implementation to verification, integration, debugging, and production support.

PROFESSIONAL EXPERIENCE

CROUSE PJS Co.

Tehran, Iran

Senior Embedded Software Engineer / Automotive Software Engineer

Sep 2021 – Present

Telematics Control Unit

- Integrated modem and telematics software on **Meig MA800** and **Quectel MC60** platforms, covering **cellular voice, SMS, data, GNSS, SIM, MQTT, HTTP, OTA/FOTA, eCall/bCall, PMU**.
- Implemented **Embedded Linux** applications and services using **processes, threads, ADB, GDB, cross-compilation, and user/kernel-space interfaces**.
- Designed low-level drivers and firmware for **CAN, UART, SPI/QSPI, I2C, I2S, USB, SDIO, GPIO, DMA, displays, SDRAM, and QSPI Flash**.
- Reduced boot time by optimizing the **U-Boot boot process**, increasing **SPI** speed, and removing/optimizing unnecessary **kernel modules and init scripts**.
- Debugged hardware/software issues and performed **root-cause analysis** using **JTAG, GDB, oscilloscopes, logic analyzers, and CAN tools**.

Automotive Cluster / Graphics Platform

- Designed **QSPI/SDRAM** drivers, custom **Flash Loader**, **linker configuration**, and **RTOS task architecture**.
- Integrated **LVGL** and **TouchGFX**, achieving stable **60 FPS** at 1024×768 through graphics optimization.
- Optimized **UI processing** with optimized **UI design techniques**.
- Resolved **tearing issues** in **LTDC** and **DMA2D** on **STM32** and **GD32** during digital cluster rendering.
- Designed and developed digital cluster software using **GUI Guider** for **LVGL**.
- Debugged hardware/software issues and performed **root-cause analysis** using **JTAG, GDB, oscilloscopes, logic analyzers, and CAN tools**.

Automotive Radio / Infotainment

- Implemented **USB, display, audio, CAN, and Bluetooth** functionality and a **Bluetooth IC production programmer**.
- Designed and optimized **USB disk scanning algorithms** for audio files.
- Designed control programs for handling/decoding **MP3, WMA, and WAV** files.
- Resolved **STM32 HAL USB stack** issue.
- Built automated **test benches** and **production-test solutions** including **CAN/MFD decoding**.
- Automated **unit, integration, system, HIL/SIL, regression, and production tests** using **Python, Unity, VectorCAST** and dedicated test benches.
- Debugged hardware/software issues and performed **root-cause analysis** using **JTAG, GDB, oscilloscopes, logic analyzers, and CAN tools**.

Rahbaran Internet of Things

Tehran, Iran

Electronics Engineer

Sep 2020 – Sep 2021

- Engineered STM32-based embedded firmware for IoT and wireless applications, integrating Zigbee and LoRa modules, including peripheral functionality for LoRa Gateway systems using STM32F030, within Linux development and Git-based workflows.

Kooreh Shahab

Tehran, Iran

Electrical Design Engineer

Apr 2019 – Sep 2020

- Designed and documented electrical schematics and PCB solutions for industrial equipment and power-supply applications, with hands-on experience in induction furnaces, industrial electrical systems, and switching power supplies.

PERSONAL PROJECTS

Lichee Pi Zero – Yocto ADB Integration

[GitHub](#)

- Developed a Yocto meta-layer for Lichee Pi Zero to add ADB over USB support.

EDUCATION

M.Sc., Electrical & Electronic Engineering – Automotive Engineering Iran University of Science and Technology

Graduated as **valedictorian** with a focus on automotive electrical/electronic systems, embedded systems, automotive software, and intelligent automotive systems.

[link](#)

RESEARCH EXPERIENCE

- Road damage detection with bounding box and generative adversarial networks based augmentation methods, *IET Image Processing*, 2024. [DOI](#)
- Improving the Accuracy of Road Damage Detectors Using Traditional Augmentations and Object Bounding Box Augmentation. [JIAEEE – Research Article](#)
- Regenerative Braking in Electric and Hybrid Vehicles – Review study of regenerative braking systems in electric and hybrid vehicles. [Civilica – Conference Paper](#)

B.Sc., Avionics Engineering

Civil Aviation Technology College

Final Project: [PLC-Based Control System](#) – PLC programming, I/O integration, and control-system testing.

TECHNICAL SKILLS

Programming: C, Embedded C, Python

Embedded Systems: ARM Cortex-M, STM32F1/F4/F7/H7, GD32F1/F4/F7/H7, RTOS, uC/OS, Bootloaders, Low-Level Drivers

Embedded Linux: Linux, Yocto, BitBake, Cross Compilation, GDB, ADB, Linux Services, Processes/Threads, User/Kernel Space

Automotive: CAN, UDS/ISO 14229, TCU, Telematics, eCall, bCall, OTA/FOTA, Cellular, MQTT

Modem / Telematics Platforms: Meig MA800, Quectel MC60

Peripherals: CAN, SPI/QSPI, I2C, UART, USB, I2S, SDIO, DMA, GPIO

Graphics: LVGL, TouchGFX, TFT-LCD, DMA2D, SDRAM

AI / Computer Vision: YOLOv5, Object Detection, OpenCV, Deep Learning, GAN, Image Processing, Data Augmentation

Testing: Unity, Integration, System, HiL/SiL, Test Automation, Regression Testing, MISRA-C, Static Analysis

Tools: makefile, Git, SVN, CANoe, IAR, Keil, STM32CubeIDE, GDB, Ozone, JTAG, VectorCAST, Robot Framework, PVS-Studio, IBM DOORS, Enterprise Architect, Confluence

COURSES & CERTIFICATES

- Advanced Hardware Design – Fedevel Academy
- Advanced PCB Layout Course – Fedevel Academy
- Altium Designer – Technical & Vocational Training Organization
- AutoCAD Electrical – Technical & Vocational Training Organization
- ARM Fundamentals with STM32 Microcontrollers – TeachTechs
- C Programming in NASA Style – TeachTechs
- English Language Diploma – Puyesh Institute

STANDARDS & METHODOLOGIES

MISRA-C, ISTQB, V-Model, Agile, Requirements Engineering, Software Verification & Validation, ASPICE concepts.

Languages: Persian – Native • English – Good Professional Working Proficiency